

CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION

INFORMATION TECHNOLOGY

HYBRID

Paper 02 – General Proficiency

2 hours

09 JANUARY 2026 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of FOUR questions. Answer ALL questions.
2. All responses **MUST** be written in the answer booklet provided.
3. Your responses **MUST** be written in English.
4. You are advised to take some time to read through the paper and plan your answers.
5. It is an offence to share your keycode with any other candidate or to log in with another candidate's details.
6. Any attempt to change the configuration of this machine, connect external devices, connect to external networks or to in any way initiate communication with resources other than the URL provided will result in your disqualification, you being shut out of the system and the cancellation of your entire test.

Answer ALL questions.

1. A new technology and security director has joined the New Technologies Centre. The new director's job is to review the centre's technology and security policies to strengthen the network and reduce threats.

- (a) State the difference between 'cybersecurity' and 'computer security'. (2 marks)
- (b) Suggest TWO examples of software solutions that the director of technology and security may implement to secure the centre's network, and state how ONE of the named software solutions would help to protect the network. (3 marks)
- (c) The director needs to explain to other members of the team some possible threats that are to be considered. Match the number representing the threat in the first column with the letter representing the description of the threat in the third column. Each letter can be used once, more than once or not at all.

	Threat		Description
(i)	Electronic eavesdropping	A	The use of technology to gather and steal personal information about an individual
(ii)	Industrial espionage	B	The use of computer systems to distribute information that is false or misleading
(iii)	Identity theft	C	Use of electronic devices to monitor communication between two or more parties on a network
(iv)	Propaganda	D	Theft of computer programs and the unauthorized use and distribution of programs
(v)	Software piracy	E	Gaining an advantage by illicitly accessing information about a competitor's marketing strategies

- (d) Consider the following database tables named Suppliers and Inventory.

Suppliers

Supplier ID	Company Name	Contact's FirstName	Contact's LastName	Address	Contact Number
TB19230	TekBiz	Lincoln	Clarke	7 Humphrey Blvd	875-9786
CW00201	Computer World	Asha	Davy	23 Kingston Street	353-6746
IN11243	Internet Needs Plus	Simon	Jones	19 Seaweed Close	332-0958
NS19431	Network Solutions	Theresa	Malah	55 Thelma Way	771-9903

Inventory

ItemNo.	ItemName	Cost (USD)	Quantity	SupplierID
329370	UPS	\$57.89	5	CW00201
112321	Norton 360 antivirus	\$34.99	23	TB19230
863823	Fireproof cabinet	\$99.99	4	TB19230
738402	Fingerprint scanner	\$63.22	10	NS194311
738404	Security camera	\$85.99	14	IN11243

(i) State the name of the primary key in the Inventory table.

(1 mark)

(ii) Complete the table design below for the Inventory table.

FieldName	DataType	Description	FieldSize
ItemNo.			6
ItemName	Text	The name of the item	
Cost (USD)		The cost of the item	
Quantity	Number		
SupplierID		The ID number of the supplier for each item	

(4 marks)

- (iii) Complete the design view extract below to show a simple query that will list ALL the items in the inventory sold by TekBiz.

Show the following fields in the query—SupplierID, ItemName, Cost and Quantity.

Field					
Table					
Sort					
Show	☐	☐	☐	☐	☐
Criteria					
Or					

(5 marks)

Total 20 marks

2. The Transport Authority of Lancaster is working on organizing driver data nationwide. You were hired as a consultant to help with the changes. Data organization and access are the top priorities.

- (a) Identify ONE difference between 'serial file organization' and 'sequential file organization'.
(2 marks)
- (b) State the appropriate file organization method for EACH of the following scenarios.

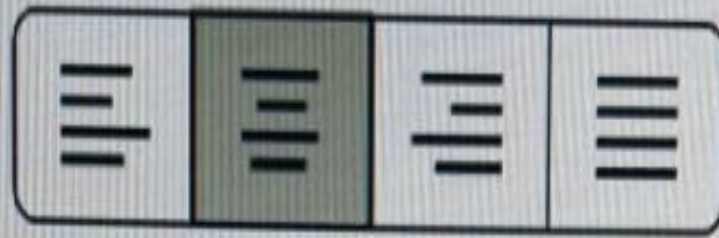
Scenario	File Organization Method
(i) Customer Service call records at the Transport Authority's call centre	
(ii) Employees' lunch orders at the staff cafeteria	
(iii) Monthly payroll information for employees	
(iv) Archived news stories for the Transport Authority, stored on a magnetic tape	

(4 marks)

- (c) The Transport Authority has hired a research specialist to conduct a review on road usage for the year.
 - (i) Identify TWO sources of information that can be used by the research specialist.
(2 marks)
 - (ii) State ONE way in which EACH of the sources of information identified in (c) (i) can be used by the Transport Authority.
(2 marks)
- (d) The research specialist hired by the Transport Authority completed a report of findings to share with the CEO. In reviewing the report, the specialist noticed that a few paragraphs in the report were not properly structured.

Write the steps required, in the correct order, that should be taken to place the fourth paragraph above the third paragraph.
(2 marks)

- (e) The research specialist used the formatting feature highlighted below in the report.



Identify the alignment feature selected and describe how text formatted using this alignment will be displayed. **(2 marks)**

- (f) A table of contents was placed at the beginning of the document to assist readers in locating sections within the report. List the steps required, in the correct order, to generate an automated table of contents in the document. **(2 marks)**
- (g) The completed traffic report was uploaded to the Transport Authority's website and accessed using a hyperlink. Define the term 'hyperlink'. **(1 mark)**
- (h) State THREE ways in which hyperlinks can be used on a website. **(3 marks)**

Total 20 marks

3. The use of information and communications technology (ICT) within the workplace has its advantages and disadvantages.

- (a) For the list of sectors in the table below, identify ONE impact of ICT and provide ONE example of how it is applied in EACH sector.

Sector	ICT impact	Example of its Application
(i) Education		
(ii) Law Enforcement		
(iii) Medicine		

(6 marks)

- (b) Define the term 'telecommuting' and state ONE advantage of telecommuting. (2 marks)
- (c) Information will be saved on a company's network both locally and online. Outline the difference between 'cloud storage' and 'local storage'. (2 marks)

- (d) The chief information officer (CIO) would like to implement a program that determines the total distance travelled to work by 50 employees as a test project. The program should
- accept the following data: employee's name, and employee vehicle details such as the make, model, year, petrol type (gas or diesel) and distance travelled to work (in kilometres).
 - calculate the total combined distance for all the employees
 - print the total combined distance for all the employees.
- (i) State any TWO variables and their data types required for the program. (2 marks)
- (ii) Outline FOUR characteristics of algorithms. (8 marks)
- (iii) Write an algorithm to create the program requested by the CIO in (d) on page 15. (5 marks)

Total 25 marks

4. (a) Cambridge Statistics Centre collects data, manipulates and presents it to help with making decisions within the city. Differentiate between the terms 'data' and 'information'.
(2 marks)

- (b) Cambridge Statistics Centre hired individuals within its IT department to assist in making tasks easier for the company. Complete the table below by inserting the name of the correct professions in the first column to match the description of the roles given in the second column.

Role	Profession
(i) Ensures that the developed application works the way it is intended to	
(ii) Ensures that connectivity within the organization is robust	
(iii) Designs, creates, and maintains data storage systems	

(3 marks)

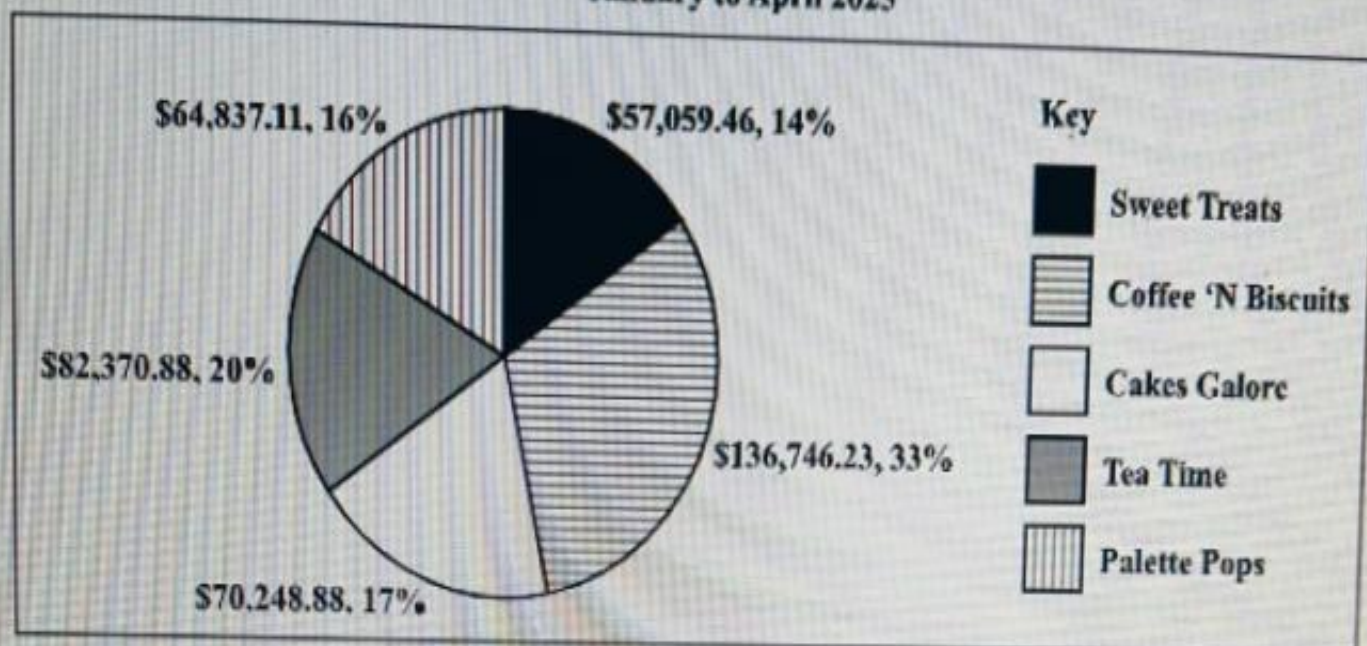
- (c) The marketing team at Cambridge Statistics Centre were tasked with identifying the leading coffee shops within the city. The spreadsheet extract below shows data for the first four months of 2025 that was collected and stored.

	A	B	C	D	E	F	G
1	Cambridge Centre Coffee Shops (2025, January to April Sales)						
2	Shops	January	February	March	April	Total	Average
3	Sweet Treats	\$23,650.45	\$11,353.56	\$11,157.00	\$10,898.45	\$57,059.46	
4	Coffee 'N Biscuits	\$23,520.23	\$27,235.22	\$51,337.78	\$34,653.00	\$136,746.23	
5	Cakes Galore	\$21,323.77	\$21,121.89	\$11,823.32	\$15,979.90	\$70,248.88	
6	Teas Time	\$21,878.34	\$10,834.00	\$17,902.10	\$31,756.44	\$82,370.88	
7	Palette Pops	\$16,054.00	\$11,502.30	\$15,681.11	\$21,599.70	\$64,837.11	

- (i) Write the function, including the range, that can be used to determine the number of shops in the spreadsheet. (2 marks)
- (ii) Write the function formula that would be required to display the correct
- total in Cell F5 (2 marks)
 - average in Cell G3. (2 marks)

- (d) A chart was required to show the percentage share/sales for each coffee shop. The results are displayed below.

Chart Showing the Percentage Share for the Cambridge Centre Coffee Shops for January to April 2025



- (i) Identify the type of chart used to display the data above. (1 mark)
- (ii) List the steps required, in the correct order, to create the chart identified in (d) (i). (2 marks)
- (iii) State the name and corresponding percentage value of the coffee shop which had the **highest** sales for the quarter. (1 mark)
- (e) Define the term "run-time error". (1 mark)
- (f) State the difference between 'pseudocode' and 'flowchart'. (2 marks)

(g) The IT Team at Cambridge Statistics Centre is required to create a program that will be used by the team to collect data from the coffee shops in Cambridge Centre for the first four months of 2025. The program should

- accept the following data: name of coffee shop, and sales for the period January to April.
- calculate the total sales for the period
- calculate the average of the sales for the period
- print the name of the coffee shop, the total sales and the sales average for the period.

(i) Complete the following defining diagram (IPO Chart) that represents the program requirements above.

Input	Processing	Output
		I

(4 marks)

Input	Processing	Output

(4 marks)

- (ii) As the programmer from the IT team, you are required to write the pseudocode algorithm to instruct the program to collect the sales data for the coffee shops. Draw the flowchart for the program described in part (g) above. (3 marks)

Total 25 marks